

SEDALIA AGCM, AB, Canada

WMO: 713990

Lat: 51.590N

Lon: 110.755W

Elev: 2477

StdP: 13.43

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.1	-19.5	-31.5	1.0	-25.0	-25.6	1.4	-19.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.5	87.4	61.4	83.7	60.8	80.0	59.8	65.3	78.2	63.5	77.2	61.9	75.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	87.9	70.5	58.9	81.3	68.2	56.9	75.7	65.9	31.5	77.7	30.1	78.0	28.9	75.4	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	DB	-32.3	94.0	3.4	4.2	-34.8	97.0	-36.8	99.5	-38.7	101.9	-41.2	105.0	(4)
(5)			WB	-32.5	69.5	3.4	2.2	-34.9	71.1	-36.9	72.4	-38.9	73.7	-41.3	75.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	12.6	10.3	24.9	38.6	51.4	58.9	64.3	62.9	53.7	39.8	24.2	10.6
	DBStd	22.58	14.81	14.15	14.18	9.76	7.73	6.07	5.29	6.03	8.59	9.27	13.13	14.78
	HDD50	5884	1160	1113	778	355	79	6	0	1	56	338	776	1222
	HDD65	10052	1625	1533	1243	793	424	201	78	112	349	782	1225	1687
	CDD50	1441	0	0	0	12	122	272	444	402	168	20	1	0
	CDD65	134	0	0	0	0	1	17	57	48	11	0	0	0
	CDH74	2596	0	0	0	9	140	312	932	859	335	9	0	0
	CDH80	788	0	0	0	0	18	79	298	278	115	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	13.0	0.6	0.3	0.5	0.8	1.5	2.9	2.1	1.7	1.0	0.6	0.5	0.4
	PrecMax	18.5	1.9	0.7	1.4	2.4	3.1	5.9	5.0	3.3	3.3	1.3	1.7	0.7
	PrecMin	6.8	0.1	0.0	0.1	0.1	0.1	0.7	0.8	0.2	0.0	0.2	0.0	0.1
	PrecStd	3.0	0.4	0.2	0.3	0.6	0.9	1.3	0.9	0.8	0.8	0.3	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	43.1	61.4	74.2	82.4	88.4	91.6	90.9	89.2	74.0	60.5	43.3
		MCWB	37.7	37.6	46.5	51.3	56.5	61.0	63.4	62.1	60.0	54.6	45.6	37.1
	2%	DB	39.3	37.3	54.6	67.7	78.0	81.4	87.5	86.6	83.2	66.4	51.8	39.1
		MCWB	35.2	33.3	42.7	48.6	54.7	59.3	63.3	61.3	58.5	51.0	41.6	35.0
	5%	DB	36.5	33.9	49.0	61.7	73.9	77.1	83.1	82.8	76.9	60.9	45.2	34.4
		MCWB	33.3	31.3	39.9	45.0	53.4	58.7	62.4	60.7	56.2	47.6	38.3	30.9
	10%	DB	33.1	30.5	42.7	56.3	69.1	73.6	79.1	78.9	71.6	55.6	40.2	29.5
		MCWB	30.3	28.4	36.4	42.6	51.0	57.2	61.5	59.8	54.3	45.1	35.2	27.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.9	38.0	47.1	51.7	58.5	65.5	69.6	66.4	61.8	56.4	47.6	38.1
		MCDB	42.4	42.4	59.4	71.2	77.6	77.8	84.9	79.3	81.2	72.1	59.6	41.9
	2%	WB	35.3	33.7	43.4	48.7	56.3	62.5	66.3	64.3	59.3	52.1	42.7	34.7
		MCDB	38.6	36.8	53.1	65.8	74.5	76.0	78.2	78.7	79.3	62.5	50.4	38.8
	5%	WB	33.3	31.5	40.3	45.8	54.4	60.5	64.3	62.7	57.2	48.6	38.9	31.2
		MCDB	36.3	34.1	48.3	59.8	70.7	73.4	77.4	77.6	74.0	58.7	44.3	34.3
	10%	WB	30.4	28.3	36.6	43.4	52.3	58.5	62.7	61.1	55.1	45.9	35.7	27.3
		MCDB	33.1	30.5	42.1	54.9	66.6	70.5	75.6	75.3	70.2	55.2	39.7	29.6
Mean Daily Temperature Range	5% DB	MDBR	18.5	19.2	20.7	23.4	25.6	23.5	25.5	25.8	26.2	23.2	18.5	17.3
		MCDBR	19.0	20.8	28.3	34.6	33.8	30.0	32.9	34.0	36.4	31.7	26.0	21.4
		MCWBR	15.5	17.6	17.2	17.3	15.2	12.9	12.5	12.8	15.4	17.5	17.2	17.5
	5% WB	MCDBR	18.7	20.4	26.9	32.1	29.8	26.0	27.4	28.6	34.2	30.1	24.3	21.0
		MCWBR	15.8	17.4	16.9	16.8	14.2	12.3	12.6	12.2	15.1	16.9	16.7	17.5
Clear-Sky Solar Irradiance	taub	0.264	0.310	0.294	0.346	0.373	0.372	0.365	0.381	0.328	0.287	0.262	0.256	
	taud	2.338	2.224	2.437	2.309	2.275	2.333	2.388	2.307	2.442	2.530	2.470	2.330	
	Ebn at Noon	242	254	290	283	279	280	280	268	273	268	246	225	
	Edh at Noon	19	29	29	36	40	38	36	36	28	21	17	16	
All-Sky Solar Radiation	RadAvg	369	707	1143	1537	1854	1920	2057	1676	1201	727	385	279	
	RadStd	35	61	87	126	124	119	89	124	110	91	32	26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page